

You are evaluating whether a reused ontology namespace in FASTO requires updating. This is a non-interactive task. Use the GPT 5.5 High Model. Return exactly one valid downloadable JSON file and nothing else.

Do not:

- ask questions or request clarification;
- output Markdown, comments, or code fences;
- explain your reasoning outside the required JSON fields;
- summarize the ontology outside the schema.

Task

Analyze the FHIR STU3 entities used in FASTO and determine whether they are unchanged, renamed, replaced, removed, split, merged, or structurally changed in FHIR R5.

Also assess FASTO-created entities that use the FHIR namespace and determine whether FHIR R5 provides corresponding entities that may replace them.

Where supported by the evidence, identify relevant FHIR R5 entities, classes, individuals, or axioms that cover the FASTO domain but were not included in the FHIR content used by FASTO.

Evidence

Four evidence sources are provided.

1. Primary evidence: The original FASTO_RDF.owl in RDF/XML syntax. This is the authoritative source.
2. Secondary evidence: A rule-based verbalizer generated from a temporary sanitized copy of FASTO. This exists only to improve readability.
3. Tertiary evidence: FHIR R5 represented as an RDF/Turtle file.

Use this source as the newer FHIR version against which the FHIR STU3 entities used in FASTO must be compared.

4. Supplementary evidence

A JSON file that distinguishes:

- entities originating from FHIR;
- entities created by FASTO while using the FHIR namespace.

Use this distinction when assessing version drift and update requirements.

Evidence rules

- RDF/XML is the primary source for determining which FHIR entities are used in FASTO.
- The verbalizer is supplementary readability evidence only.
- Do not compare the RDF/XML and verbalizer as separate ontology versions.

- Use the FHIR R5 RDF/Turtle file to compare old and new entities, classes, individuals, and axioms.
- Treat FASTO-created entities in the FHIR namespace separately from entities originating from FHIR.
- Consider whether a FASTO-created entity should be updated when FHIR R5 provides a corresponding entity.
- Report only updates and conclusions supported by the supplied evidence.
- Do not invent ontology contents, entity ownership, mappings, versions, import histories, or intended semantics.
- Mark uncertain mappings as requiring expert review.
- Recommend relevant FHIR R5 entities, classes, individuals, or axioms when the evidence shows that they cover the FASTO domain and were not included in the FHIR STU3 content used by FASTO.

Reporting rules

- Return exactly one valid JSON object.
- Use confidence values from 0.0 to 1.0.
- Use unclear or no_clear_mapping when the evidence does not support a definite mapping.
- Use manual_expert_review when the mapping or update decision remains uncertain.

Field rules

mapping_status: Select exactly one:

- unchanged
- renamed
- replaced
- removed
- split
- merged
- structurally_changed
- no_clear_mapping
- locally_extended
- unclear

update_recommendation: Select exactly one:

- keep
- replace
- review
- remove
- split
- manual_expert_review

old_entity_type: Select exactly one:

- class

- object_property
 - data_property
 - individual
 - axiom
 - unknown
- severity: Select exactly one:
- error
 - warning
 - info

Required JSON schema

```
{
  "task": "reused_ontology_version_drift_analysis",
  "model": "",
  "input_condition": "IIm_only",
  "target_reused_ontology": {
    "name": "FHIR",
    "old_version": "STU3",
    "new_version": "R5",
    "old_namespace": "",
    "new_namespace": ""
  },
  "term_assessments": [
    {
      "old_entity": "",
      "old_entity_type": "unknown",
      "old_label": "",
      "old_definition": "",
      "new_candidate_entity": "",
      "new_candidate_label": "",
      "mapping_status": "unclear",
      "evidence": "",
      "update_recommendation": "manual_expert_review",
      "severity": "warning",
      "confidence": 0.0,
      "additional_comment": ""
    }
  ],
  "summary": {
    "number_of_terms_reviewed": 0,
    "number_requiring_update": 0,
    "number_requiring_manual_review": 0,
    "main_update_risks": []
  }
}
```

```
},  
"global_comments": [  
  {  
    "comment": "",  
    "evidence": "",  
    "confidence": 0.0  
  }  
]  
}
```

Internal quality checks

Before returning the JSON, internally verify that:

- every assessed STU3 entity is supported by the FASTO evidence;
- every proposed R5 mapping is supported by the FHIR R5 evidence;
- FASTO-created entities using the FHIR namespace are distinguished from FHIR-origin entities;
- uncertain mappings are marked for expert review;
- update recommendations are supported by evidence;
- no FHIR R5 entities were invented;
- all required fields and arrays are present;
- the summary counts match the term assessments;
- the final output is valid JSON

Now analyze the supplied evidence and return the downloadable JSON file only.